COHOMOLOGY ON X AGREES WITH COHOMOLOGY ON THE LISSE-ETALE AND FLAT-FPPF SITES

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Allowed standard prerequisites (editorial).

- Cohomology on subcategories. Sheaves on Stacks, Lemma stacks-sheaves-lemma-cohomology-on-subcategory. Both parts.

Source reference. Original lemma 0788: sources/stacks-chow/tags.txt, source lines 9228–9228. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 07AK: sources/stacks-chow/tags.txt, source lines 9310–9310. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack. Notation as in Lemma 0788.*

(1) *For an abelian sheaf $\mathcal{F}$ on $\mathcal{X}_{étale}$ we have*
   (a) *$H^p(\mathcal{X}_{étale}, \mathcal{F}) = H^p(\mathcal{X}_{lisse,étale}, g^{-1}\mathcal{F})$, and*
   (b) *$H^p(x, \mathcal{F}) = H^p(\mathcal{X}_{lisse,étale}/x, g^{-1}\mathcal{F})$ for any object x of $\mathcal{X}_{lisse,étale}$.*
   The same holds for sheaves of modules.

(2) *For an abelian sheaf $\mathcal{F}$ on $\mathcal{X}_{fppf}$ we have*
   (a) *$H^p(\mathcal{X}_{fppf}, \mathcal{F}) = H^p(\mathcal{X}_{flat,fppf}, g^{-1}\mathcal{F})$, and*
   (b) *$H^p(x, \mathcal{F}) = H^p(\mathcal{X}_{flat,fppf}/x, g^{-1}\mathcal{F})$ for any object x of $\mathcal{X}_{flat,fppf}$.*
   The same holds for sheaves of modules.

Proof. Part (1)(a) follows from Sheaves on Stacks, Lemma 07AK applied to the inclusion functor $\mathcal{X}_{lisse,étale} \to \mathcal{X}_{étale}$. Part (1)(b) follows from part (1)(a). Namely, if x lies over the scheme U, then the site $\mathcal{X}_{étale}/x$ is equivalent to $(Sch/U)_{étale}$ and $\mathcal{X}_{lisse,étale}$ is equivalent to $U_{lisse,étale}$. Part (2) is proved in the same manner. □

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